Lecture 11

Fourier analysis and convolution

Objectives. *On the circle, a function has scalar Fourier coefficients. On a nonabelian compact group, the coefficient at an irreducible representation Π is an operator $\hat{f}(\Pi)$. We derive Plancherel from Peter–Weyl. We compute how convolution acts on these operators. We identify the convolution operators that commute with all translations.*

By the end of this lecture you will be able to:

- Compute $\hat{f}(\Pi)$ for matrix coefficients and derive the Plancherel theorem from Peter–Weyl.
- Prove $(f * h)^{\wedge} = \hat{h}\,\hat{f}$ and check this order by a direct $\mathrm{SU}(2)$ computation.
- Separate L^2 convergence of spectral partial sums from pointwise or uniform convergence.
- Characterise central convolution operators and write the isotypic projectors.

11.1 The operator-valued transform

Throughout, K is a compact Lie group with normalised Haar measure dg, so $\int_K dg = 1$. From Lecture 5 we import the following facts. The measure is invariant under left translation, right translation and inversion $g \mapsto g^{-1}$. A continuous function $\varphi \geq 0$ with $\int_K \varphi\, dg = 0$ vanishes. This property is strict positivity, which we call *positivity* below. We also import Fubini's theorem for continuous functions on $K \times K$ from Bröcker–tom Dieck, Section I.5, as in Lecture 10.

Every finite-dimensional representation is unitarisable and is a direct sum of irreducible ones. Finally, Schur orthogonality holds in the following form. For irreducible unitary representations Π, Π', vectors $u, v \in V_\Pi$ and vectors $u', v' \in V_{\Pi'}$,

$$\int_K \overline{\langle u, \Pi(g)v\rangle}\, \langle u', \Pi'(g)v'\rangle\, dg = \begin{cases} d_\Pi^{-1}\langle u', u\rangle\langle v, v'\rangle, & \Pi' = \Pi, \\ 0, & \Pi' \not\cong \Pi, \end{cases} \tag{11.1}$$

where $d_\Pi = \dim V_\Pi$.

From Lecture 10 we import two facts. The space $L^2(K)$ is the Hilbert completion of $C(K)$ with respect to the inner product $\langle \varphi, \psi\rangle = \int_K \overline{\varphi}\,\psi\, dg$. By the Peter–Weyl theorem, the span $\mathscr{R}(K)$ of matrix coefficients of finite-dimensional representations is dense in $L^2(K)$.

Fix a set $\widehat{K}$ containing exactly one irreducible unitary representation (Π, V_Π) from each equivalence class. On $\mathrm{End}(V_\Pi)$ we use the Hilbert–Schmidt inner product $\langle A, B\rangle_{\mathrm{HS}} = \mathrm{tr}(A^* B)$. For $u, v \in V_\Pi$, the operator uv^* is given by $w \mapsto \langle v, w\rangle u$.

Definition 11.1. Let $f \in C(K)$ and $\Pi \in \widehat{K}$. The *Fourier transform* of f at Π is

$$\hat{f}(\Pi) = \int_K f(g)\, \Pi(g^{-1})\, dg \in \mathrm{End}(V_\Pi),$$

integrated entry by entry in any basis. For $A \in \mathrm{End}(V_\Pi)$ the *matrix coefficient* c_A is the function given by $c_A(g) = \mathrm{tr}(A\,\Pi(g))$. The space of matrix coefficients of Π is $M_\Pi = \{c_A : A \in \mathrm{End}(V_\Pi)\}$.

For $K = \mathrm{U}(1)$ and $\Pi = \chi_n$ this is the classical coefficient $\frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta}) e^{-in\theta}\, d\theta$. The inverse g^{-1} in the definition gives the minus sign.

A rank-one operator gives the familiar coefficient $c_{uv^*}(g) = \langle v, \Pi(g)u\rangle$. Let Π' be equivalent to Π, say $\Pi'(g) = T\Pi(g)T^{-1}$ with T invertible. Then

$$\mathrm{tr}\big(A\,\Pi'(g)\big) = \mathrm{tr}\big(T^{-1}AT\,\Pi(g)\big) = c_{T^{-1}AT}(g).$$

Hence the matrix coefficients of Π' also lie in M_Π. So M_Π does not depend on the choice of representative. By complete reducibility, $\mathscr{R}(K)$ is the algebraic sum of the spaces M_Π.

These objects are those of Lecture 10 under other names. There $\widehat{K}$ is $\mathrm{Irr}(K)$, the space M_Π is E_Π, and $c_A = d_\Pi^{-1/2}\Phi_\Pi(A)$. The operator uv^* is $u\langle v, \cdot\rangle$, so the letters u and v are exchanged. The Peter–Weyl theorem already gives that the spaces M_Π are orthogonal and that $L^2(K)$ is their Hilbert sum. It also gives $\|f\|^2 = \sum_\Pi \|A_\Pi\|_{\mathrm{HS}}^2$ when $f = \sum_\Pi \Phi_\Pi(A_\Pi)$, and the coefficient lemma below gives $A_\Pi = \sqrt{d_\Pi}\,\hat{f}(\Pi)$. So what is new in this lecture is the formula for the projection and the operator form of the coefficient.

Lemma 11.2. *For $f \in C(K)$, $\Pi \in \widehat{K}$ and $B \in \mathrm{End}(V_\Pi)$,*

$$\langle c_B, f\rangle = \langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}}. \tag{11.2}$$

Proof. For every operator X we have $\overline{\mathrm{tr}\, X} = \mathrm{tr}\, X^*$. Since $\Pi(g)^* = \Pi(g^{-1})$,

$$\overline{c_B(g)} = \mathrm{tr}\big(\Pi(g)^* B^*\big) = \mathrm{tr}\big(B^*\,\Pi(g^{-1})\big).$$

Multiply by $f(g)$ and integrate. The trace is linear, so it passes through the entrywise integral. This gives $\mathrm{tr}(B^*\hat{f}(\Pi))$. ∎

We use (11.2) to extend the transform to $L^2(K)$. The left side of (11.2) is L^2-continuous in f. For fixed f, it is an antilinear functional of B. Hence the finite-dimensional Riesz theorem gives a unique operator $\hat{f}(\Pi)$ satisfying (11.2). For continuous f, this operator agrees with the integral of Definition 11.1, by the pairing lemma, Lemma 11.2.

Lemma 11.3. *Let $\Pi, \Pi' \in \widehat{K}$, $A \in \mathrm{End}(V_\Pi)$ and $B \in \mathrm{End}(V_{\Pi'})$. Then*

$$\langle c_B, c_A\rangle = \begin{cases} d_\Pi^{-1}\langle B, A\rangle_{\mathrm{HS}}, & \Pi' = \Pi, \\ 0, & \Pi' \neq \Pi. \end{cases} \tag{11.3}$$

Consequently $\hat{c}_A(\Pi) = d_\Pi^{-1} A$, and $\hat{c}_A(\Pi') = 0$ for $\Pi' \neq \Pi$.

Proof. Both sides are linear in A and antilinear in B. Rank-one operators span, so take $A = uv^*$ and $B = u'v'^*$. Distinct elements of $\widehat{K}$ are inequivalent. By (11.1),

$$\langle c_B, c_A\rangle = \int_K \overline{\langle v', \Pi'(g)u'\rangle}\,\langle v, \Pi(g)u\rangle\, dg = \delta_{\Pi\Pi'}\, d_\Pi^{-1}\langle v, v'\rangle\langle u', u\rangle.$$

On the other hand $(u'v'^*)^* = v'u'^*$, so

$$\langle B, A\rangle_{\mathrm{HS}} = \operatorname{tr}\big(v'u'^*uv^*\big) = \langle u', u\rangle\operatorname{tr}(v'v^*) = \langle u', u\rangle\langle v, v'\rangle.$$

This proves (11.3). By (11.2), $\langle B, \hat{c}_A(\Pi')\rangle_{\mathrm{HS}} = \langle c_B, c_A\rangle$ for every B. Comparing with (11.3) gives the transforms. ∎

11.2 Plancherel from Peter–Weyl

For $f \in L^2(K)$ and a finite set $F \subset \widehat{K}$, the *spectral partial sum* is

$$S_F f(g) = \sum_{\Pi\in F} d_\Pi\, \operatorname{tr}\big(\hat{f}(\Pi)\,\Pi(g)\big).$$

Theorem 11.4 (Plancherel). *Let $f, h \in L^2(K)$.*

(a) The orthogonal projection of $L^2(K)$ onto M_Π is $P_\Pi f = d_\Pi\, c_{\hat{f}(\Pi)}$.

(b) For every $\varepsilon > 0$ there is a finite $F_0 \subset \widehat{K}$ with $\|f - S_F f\| < \varepsilon$ for every finite $F \supseteq F_0$.

(c) $\|f\|^2 = \sum_\Pi d_\Pi \|\hat{f}(\Pi)\|_{\mathrm{HS}}^2$ and $\langle f, h\rangle = \sum_\Pi d_\Pi \langle \hat{f}(\Pi), \hat{h}(\Pi)\rangle_{\mathrm{HS}}$.

(d) If $\hat{f}(\Pi) = 0$ for every Π, then $f = 0$.

Proof. Step 1. By (11.3), $M_\Pi \perp M_{\Pi'}$ for $\Pi \neq \Pi'$, and $\|c_A\|^2 = d_\Pi^{-1}\|A\|_{\mathrm{HS}}^2$. Each M_Π is finite-dimensional, hence closed.

Step 2. Put $f_0 = d_\Pi c_{\hat{f}(\Pi)} \in M_\Pi$. For every $B \in \operatorname{End}(V_\Pi)$, (11.2) and (11.3) give

$$\langle c_B, f - f_0\rangle = \langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}} - d_\Pi\, d_\Pi^{-1}\langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}} = 0.$$

So $f - f_0 \perp M_\Pi$, that is, $f_0 = P_\Pi f$. Step 1 gives $\|P_\Pi f\|^2 = d_\Pi\|\hat{f}(\Pi)\|_{\mathrm{HS}}^2$.

Step 3. This step uses the Peter–Weyl theorem. Let $M_F = \bigoplus_{\Pi\in F} M_\Pi$. By Steps 1 and 2, $S_F = \sum_{\Pi\in F} P_\Pi$ is the orthogonal projection onto M_F. Hence $S_F f$ is the point of M_F nearest to f. By the Peter–Weyl theorem, there is $\varphi \in \mathscr{R}(K)$ with $\|f - \varphi\| < \varepsilon$. Since $\mathscr{R}(K)$ is the algebraic sum of the spaces M_Π, there is a finite F_0 with $\varphi \in M_{F_0}$. If $F \supseteq F_0$, then $\varphi \in M_F$, so $\|f - S_F f\| \le \|f - \varphi\| < \varepsilon$.

Step 4. Pythagoras and Step 2 give

$$\|f\|^2 = \|S_F f\|^2 + \|f - S_F f\|^2, \qquad \|S_F f\|^2 = \sum_{\Pi\in F} d_\Pi\|\hat{f}(\Pi)\|_{\mathrm{HS}}^2.$$

Hence every finite partial sum of the series in (c) is at most $\|f\|^2$. By Step 3, the partial sum over F exceeds $\|f\|^2 - \varepsilon^2$ once $F \supseteq F_0$. A sum of nonnegative terms is the supremum of its finite partial sums, so the norm identity holds. Polarisation gives the inner product identity. Finally, if $\hat{f}(\Pi) = 0$ for every Π, then the norm identity gives $\|f\| = 0$. This is the uniqueness statement (d). ∎

On $\mathrm{U}(1)$ we have $d_\Pi = 1$, and the norm identity is Parseval's identity.

Corollary 11.5 (Inversion for finite coefficient sums)**.** *Let $f \in \mathscr{R}(K)$. Then $\hat f(\Pi) \neq 0$ for only finitely many Π, and for every $g \in K$*

$$f(g) = \sum_{\Pi \in \widehat{K}} d_\Pi \operatorname{tr}\big(\hat f(\Pi)\Pi(g)\big).$$

Proof. Write $f = \sum_{\Pi \in F} c_{A_\Pi}$ with F finite. By the coefficient lemma, Lemma 11.3, $\hat f(\Pi) = d_\Pi^{-1} A_\Pi$ for $\Pi \in F$ and $\hat f(\Pi) = 0$ otherwise. Substituting returns $\sum_{\Pi \in F} c_{A_\Pi}(g) = f(g)$. ∎

For general $f \in L^2(K)$ we only have the L^2 convergence of spectral partial sums in the Plancherel theorem, Theorem 11.4(b). This statement says nothing about the value of f at a single point.

11.3 Convolution reverses the order

For a finite group G, the product of the group algebra $\mathbb{C}[G]$ (Finite Groups) is

$$\Big(\sum_{y} f(y)y\Big)\Big(\sum_{z} h(z)z\Big) = \sum_{x}\Big(\sum_{y} f(y)h(y^{-1}x)\Big)x.$$

The next definition is the same formula, with the sum over y replaced by the normalised Haar integral. Finite Groups also proved $\mathbb{C}[G] \cong \bigoplus_U \operatorname{End}(U)$, by sending $\sum f(g)g$ to the operators $\sum f(g)U(g)$. Under the transform, a function becomes a family of matrices, one for each irreducible, and convolution multiplies these matrices one block at a time. The transform $\hat f$ uses $\Pi(g^{-1})$, so it reverses the order.

Definition 11.6. For $f, h \in C(K)$ the *convolution* $f * h$ is

$$(f * h)(x) = \int_K f(y)\, h(y^{-1}x)\, dy.$$

Fix x. The map $y \mapsto y^{-1}x$ is inversion followed by right translation, and both preserve dy. So $y \mapsto h(y^{-1}x)$ has L^2 norm $\|h\|$, and Cauchy–Schwarz gives $|(f * h)(x)| \le \|f\|\,\|h\|$.

The function $f * h$ is also continuous. By the uniform-continuity lemma of Lecture 10, h is uniformly continuous on the compact group K. In other words, for every $\varepsilon > 0$ there is a neighbourhood N of e with $|h(z) - h(zw)| < \varepsilon$ for all $z \in K$ and $w \in N$. Hence $|(f * h)(x) - (f * h)(xw)| \le \varepsilon \int_K |f|\, dy$ for $w \in N$.

Theorem 11.7 (Convolution theorem)**.** *For $f, h \in C(K)$ and $\Pi \in \widehat{K}$,*

$$(f * h)^{\wedge}(\Pi) = \hat h(\Pi)\, \hat f(\Pi).$$

Proof. Exchange the iterated integrals entrywise:

$$(f * h)^{\wedge}(\Pi) = \int_K f(y)\Big(\int_K h(y^{-1}x)\, \Pi(x^{-1})\, dx\Big)dy.$$

In the inner integral substitute $x = yz$. Left invariance gives $dx = dz$, and $x^{-1} = z^{-1}y^{-1}$. So the inner integral is

$$\int_K h(z)\,\Pi(z^{-1})\Pi(y^{-1})\,dz = \hat{h}(\Pi)\,\Pi(y^{-1}).$$

The factor $\Pi(y^{-1})$ coming from f stands on the right, because $x^{-1} = z^{-1}y^{-1}$. Integrating against $f(y)$ and using the definition of $\hat{f}(\Pi)$ gives $\hat{h}(\Pi)\hat{f}(\Pi)$. ∎

Some texts use $\tilde{f}(\Pi) = \int_K f(g)\Pi(g)\,dg$. The same computation then gives $(f * h)^{\sim} = \tilde{f}\,\tilde{h}$. Check the convention before importing formulas.

Corollary 11.8. *Let $A, B \in \mathrm{End}(V_\Pi)$. Then $c_A * c_B = d_\Pi^{-1} c_{BA}$. If instead $B \in \mathrm{End}(V_{\Pi'})$ with $\Pi' \neq \Pi$, then $c_A * c_B = 0$.*

Proof. By the coefficient lemma and the convolution theorem, Lemma 11.3 and Theorem 11.7, $(c_A * c_B)^\wedge(\Pi) = d_\Pi^{-2} BA$. This is the transform of $d_\Pi^{-1} c_{BA}$. At every other element of $\widehat{K}$ both transforms vanish. The difference φ of the two sides is continuous, and $\|\varphi\| = 0$ by the uniqueness statement of the Plancherel theorem, Theorem 11.4(d). Positivity applied to $|\varphi|^2$ gives $\varphi = 0$. In the second case one of $\hat{c}_A, \hat{c}_B$ vanishes at each element of $\widehat{K}$, so the same argument gives $c_A * c_B = 0$. ∎

For rank-one operators, $(u'v'^*)(uv^*) = \langle v', u\rangle\, u'v^*$. Hence the coefficient-convolution corollary, Corollary 11.8, gives

$$\langle v, \Pi(\cdot)u\rangle * \langle v', \Pi(\cdot)u'\rangle = d_\Pi^{-1}\langle v', u\rangle\,\langle v, \Pi(\cdot)u'\rangle.$$

Fix an orthonormal basis $e_1, \dots, e_{d_\Pi}$ of V_Π. Write $E_{ij} = e_i e_j^*$ for the matrix unit. Its matrix in this basis has 1 in row i and column j, and 0 elsewhere. By the last display, the matrix coefficients $D_{ij}(g) = \langle e_i, \Pi(g)e_j\rangle$ satisfy $D_{ij} * D_{kl} = d_\Pi^{-1}\delta_{jk} D_{il}$. So the functions multiply in the natural order. By the coefficient lemma, Lemma 11.3, their transforms are the transposed matrix units $d_\Pi^{-1} E_{ji}$, and these multiply in reverse.

Example 11.9 (Direct check on SU(2))**.** Write elements of SU(2) as

$$U = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

Let Π be the defining representation on $\mathbb{C}^2$, with standard basis e_1, e_2, so $d_\Pi = 2$. Put $f(U) = U_{21} = b$ and $h(U) = U_{12} = -\bar{b}$. Since $\mathrm{tr}(E_{12}U) = U_{21}$ and $\mathrm{tr}(E_{21}U) = U_{12}$, we have $f = c_{E_{12}}$ and $h = c_{E_{21}}$. The convolution theorem predicts the transform of $f * h$. The coefficient-convolution corollary, Corollary 11.8, then gives the function, since $c_{E_{12}} * c_{E_{21}} = \tfrac{1}{2} c_{E_{21}E_{12}} = \tfrac{1}{2} c_{E_{22}}$. Hence

$$(f * h)^\wedge(\Pi) = \hat{h}\hat{f} = \tfrac{1}{4} E_{21}E_{12} = \tfrac{1}{4} E_{22}, \qquad \text{so} \quad (f * h)(U) = \tfrac{1}{2} U_{22} = \tfrac{1}{2}\bar{a}.$$

The opposite order would give the transform $\tfrac{1}{4}E_{11}$ and hence the function $\tfrac{1}{2}a$.

Now compute directly. Since $a = \langle e_1, Ue_1\rangle$ and $b = \langle e_2, Ue_1\rangle$, (11.1) gives

$$\int_{\mathrm{SU}(2)} |b|^2\,dU = \tfrac{1}{2}, \qquad \int_{\mathrm{SU}(2)} \bar{a}\,b\,dU = \tfrac{1}{2}\langle e_2, e_1\rangle = 0.$$

Let y have entries a, b and let x have entries α, β in the same pattern. Since y is unitary, $y^{-1} = y^{*}$. Hence

$$y^{-1}x = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix} \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} = \begin{pmatrix} \bar{a}\alpha + \bar{b}\beta & \bar{b}\bar{\alpha} - \bar{a}\bar{\beta} \\ -b\alpha + a\beta & a\bar{\alpha} + b\bar{\beta} \end{pmatrix}.$$

Since $h(y^{-1}x)$ is the $(1,2)$ entry of this matrix, we get

$$(f * h)(x) = \int b\,(\bar{b}\bar{\alpha} - \bar{a}\bar{\beta})\,dy = \bar{\alpha}\int |b|^2\,dy - \bar{\beta}\int \bar{a}\,b\,dy = \tfrac{1}{2}\bar{\alpha}.$$

This matches the prediction. Using $(y^{-1}x)_{21} = -b\alpha + a\beta$, the same method gives $(h * f)(x) = \tfrac{1}{2}\alpha$. So $f * h \neq h * f$.

11.4 Pointwise and uniform inversion

Lecture 10 also shows that $\mathscr{R}(K)$ is uniformly dense in $C(K)$. This density does not say that the approximants are the partial sums $S_F f$. Uniform convergence of $S_F f$ needs an extra hypothesis.

Proposition 11.10. *Let $f \in C(K)$. Assume*

$$\sum_{\Pi \in \widehat{K}} d_\Pi^{3/2}\, \|\hat{f}(\Pi)\|_{\mathrm{HS}} < \infty.$$

Then the series

$$\sum_{\Pi \in \widehat{K}} d_\Pi\, \operatorname{tr}\big(\hat{f}(\Pi)\,\Pi(g)\big)$$

converges absolutely and uniformly on K. Its sum is $f(g)$ for every $g \in K$.

Proof. We need two facts about the Hilbert–Schmidt norm. The first is Cauchy–Schwarz for $\langle \cdot, \cdot \rangle_{\mathrm{HS}}$, applied to X^{*} and Y. The second uses unitarity of $\Pi(g)$:

$$|\operatorname{tr}(XY)| \le \|X\|_{\mathrm{HS}}\,\|Y\|_{\mathrm{HS}}, \qquad \|\Pi(g)\|_{\mathrm{HS}}^2 = \operatorname{tr}\big(\Pi(g)^{*}\Pi(g)\big) = d_\Pi.$$

Hence, for every $g \in K$,

$$\big|d_\Pi \operatorname{tr}\big(\hat{f}(\Pi)\Pi(g)\big)\big| \le b_\Pi, \qquad b_\Pi = d_\Pi^{3/2}\|\hat{f}(\Pi)\|_{\mathrm{HS}}.$$

The numbers b_Π have finite sum by hypothesis. Hence, for each n, only finitely many b_Π exceed $1/n$, so only countably many terms are nonzero. By the Weierstrass test the series converges absolutely and uniformly to a continuous function Φ.

Explicitly, given $\varepsilon > 0$, choose a finite $F_1 \subset \widehat{K}$ with $\sum_{\Pi \notin F_1} b_\Pi < \varepsilon$. Then for every finite $F \supseteq F_1$,

$$\sup_{g \in K} \big|\Phi(g) - S_F f(g)\big| \le \sum_{\Pi \notin F} b_\Pi < \varepsilon.$$

Since dg has mass one, $\|\Phi - S_F f\| < \varepsilon$. Enlarge F to contain the set F_0 given for this ε by the L^2 convergence statement of the Plancherel theorem, Theorem 11.4(b). By the triangle inequality, $\|\Phi - f\| < 2\varepsilon$. Since $\varepsilon > 0$ was arbitrary, $\|\Phi - f\| = 0$. Positivity applied to $|\Phi - f|^2$ then gives $\Phi = f$. ∎

Remark 11.11 (Cited results). (i) Let K be compact and connected, with maximal torus T of dimension r. Fix a Weyl-invariant inner product on $\mathfrak{t}^*$. By Lecture 8, each $\Pi \in \widehat{K}$ has a highest weight λ_Π. Distinct Π have distinct highest weights, and these lie in the weight lattice of T, a lattice of rank r. Now let $f \in C^\infty(K)$. Then for every N there is C_N with

$$\|\hat{f}(\Pi)\|_{\mathrm{HS}} \le C_N \, (1 + |\lambda_\Pi|)^{-N} \qquad \text{for all } \Pi \in \widehat{K}.$$

We cite this from M. Sugiura, *Fourier series of smooth functions on compact Lie groups*, Osaka J. Math. 8 (1971), 33–47. The proof uses the Casimir operator.

The Weyl dimension formula, imported from the prerequisite Lie algebras course, gives $d_\Pi \le C(1 + |\lambda_\Pi|)^q$, where q is the number of positive roots. A lattice of rank r has $O(R^r)$ points of norm at most R, so $\sum_\Pi (1 + |\lambda_\Pi|)^{-M} < \infty$ for $M > r$. Taking $N > \frac{3}{2}q + r$ gives the hypothesis of the uniform inversion proposition, Proposition 11.10. So the Fourier series of a smooth function on a compact connected Lie group converges absolutely and uniformly to it.

(ii) On $K = \mathrm{U}(1)$ there are continuous functions whose symmetric partial sums diverge at a point. This is du Bois-Reymond's classical example (Grafakos, *Classical Fourier Analysis*, Section 3.3.3, Proposition 3.3.5). So convergence at every point, and hence uniform convergence, can fail for continuous functions.

11.5 Central kernels and isotypic projectors

Convolution acts one block at a time. So we can read a function through the matrices $\hat{f}(\Pi)$. Figure 11.1 draws these matrices for SU(2), including two kernels that this section computes: $k(U) = |\operatorname{tr} U|^2$ of Example 11.15, and the projector kernel e_Π of Corollary 11.14 for $\Pi = V_2$. A central kernel turns out to have scalar blocks.

Left and right translation are given, as in Lecture 10, by $(L_g f)(x) = f(g^{-1}x)$ and $(R_g f)(x) = f(xg)$. The character of Π is $\chi_\Pi = \operatorname{tr} \circ \Pi = c_I$. Unitarity gives $\chi_\Pi(g^{-1}) = \overline{\chi_\Pi(g)}$.

Definition 11.12. A function $k \in C(K)$ is *central*, or a *class function* (Lecture 9), if $k(gxg^{-1}) = k(x)$ for all $g, x \in K$. The *convolution operator* of $k \in C(K)$ is the map $T_k \colon C(K) \to C(K)$ given by $T_k f = f * k$. The operator T_k is *central* if k is central.

Substituting $y = gy'$ in the convolution integral shows that $T_k L_g = L_g T_k$ for every k. The next theorem shows that commutation with right translations characterises the central kernels.

Theorem 11.13. *For $k \in C(K)$ the following are equivalent.*

(a) k is central.

(b) $T_k R_g = R_g T_k$ for every $g \in K$.

(c) Every $\hat{k}(\Pi)$ is a scalar multiple of the identity.

In this case $\hat{k}(\Pi) = \kappa_\Pi I$ with $\kappa_\Pi = d_\Pi^{-1} \langle \chi_\Pi, k \rangle$. For $f \in C(K)$ we have $P_\Pi(f * k) = \kappa_\Pi P_\Pi f$. Hence, for finite $F \subset \widehat{K}$, the sums $\sum_{\Pi \in F} \kappa_\Pi P_\Pi f$ converge to $f * k$ in L^2, in the sense of the convergence statement of the Plancherel theorem, Theorem 11.4(b).

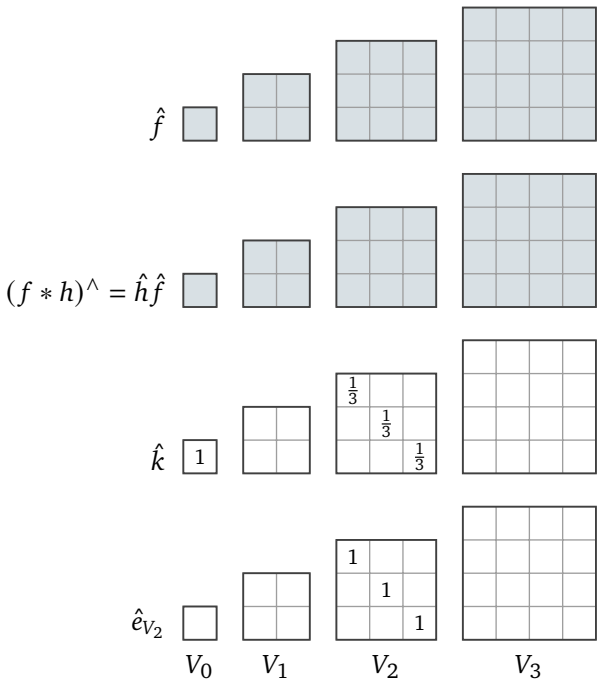


Figure 11.1: The operator blocks $\hat f(V_m)$ on SU(2), for $m = 0, 1, 2, 3$, with a shaded cell for a general entry and a blank cell for zero. The rows show a general f, a convolution $f * h$, the kernel $k(U) = |\operatorname{tr} U|^2$, and the projector kernel e_{V_2}. These are matrices of operators, not the arrays of functions of Lecture 10: since $\hat D_{ij} = d_\Pi^{-1} E_{ji}$, the two grids differ by a transpose.

Proof. *(a)⇔(c).* Fix g and put $k^g(x) = k(gxg^{-1})$. Substitute $x' = gxg^{-1}$, which preserves dx by two-sided invariance:

$$(k^g)^\wedge(\Pi) = \int_K k(x')\,\Pi(g^{-1}x'^{-1}g)\,dx' = \Pi(g)^{-1}\hat k(\Pi)\Pi(g).$$

If k is central, then $k^g = k$. Hence $\hat k(\Pi)$ commutes with every $\Pi(g)$, and by Schur's lemma it is a scalar multiple of the identity. Conversely, if every $\hat k(\Pi)$ is scalar, then k^g and k have equal transforms. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), and positivity, $k^g = k$.

(a)⇔(b). Right invariance with $y = y'g^{-1}$ gives

$$(T_k R_g f)(x) = \int_K f(y)\,k(gy^{-1}x)\,dy, \qquad (R_g T_k f)(x) = \int_K f(y)\,k(y^{-1}xg)\,dy.$$

If k is central, then $k(y^{-1}xg) = k(gy^{-1}x)$, which is (b). Conversely, assume (b) and set $x = e$. Put $\psi(y) = k(y^{-1}g) - k(gy^{-1})$. Then $\int_K f\psi\,dy = 0$ for all f. Taking $f = \bar\psi$ and using positivity gives $\psi = 0$. So $k(zg) = k(gz)$ for all z. Replacing z by zg^{-1} gives $k(z) = k(gzg^{-1})$.

Final claims. We have $\operatorname{tr}\hat k(\Pi) = \int_K k(g)\overline{\chi_\Pi(g)}\,dg = \langle \chi_\Pi, k\rangle$. Since $\hat k(\Pi) = \kappa_\Pi I$, this trace is $d_\Pi\kappa_\Pi$, which gives the formula for κ_Π. By the convolution theorem, Theorem 11.7, $(f * k)^\wedge(\Pi) = \kappa_\Pi \hat f(\Pi)$. Then the projection formula of the Plancherel theorem, Theorem 11.4(a), gives $P_\Pi(f * k) = \kappa_\Pi P_\Pi f$. So $S_F(f * k) = \sum_{\Pi\in F}\kappa_\Pi P_\Pi f$, and the L^2 convergence statement, Theorem 11.4(b), applied to $f * k$ gives the convergence. ∎

Corollary 11.14 (Isotypic projectors). *Let $e_\Pi = d_\Pi \chi_\Pi$. For $f \in C(K)$,*

$$P_\Pi f = f * e_\Pi = e_\Pi * f, \qquad (P_\Pi f)(x) = d_\Pi \int_K f(y)\, \chi_\Pi(y^{-1}x)\, dy.$$

Moreover $e_\Pi * e_{\Pi'} = \delta_{\Pi\Pi'} e_\Pi$.

Proof. First, characters are central, and $e_\Pi = d_\Pi c_I$. By the coefficient lemma, Lemma 11.3, $\hat{e}_\Pi(\Pi) = I$ and $\hat{e}_\Pi(\Pi') = 0$ for $\Pi' \neq \Pi$. So the central kernel theorem, Theorem 11.13, applies to $k = e_\Pi$ with $\kappa_{\Pi'} = \delta_{\Pi\Pi'}$. Its sums therefore equal $P_\Pi f$ once $\Pi \in F$, and they converge to $f * e_\Pi$ in L^2. Hence $\|f * e_\Pi - P_\Pi f\| = 0$. Both functions are continuous, so positivity gives $f * e_\Pi = P_\Pi f$.

Next, by the convolution theorem, Theorem 11.7, the transforms of $f * e_\Pi$ and $e_\Pi * f$ agree, because $\hat{e}_\Pi$ is scalar at every element of $\widehat{K}$. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), the difference of these functions is 0 in $L^2(K)$. Both functions are continuous, so positivity gives $f * e_\Pi = e_\Pi * f$.

Finally, the coefficient-convolution corollary, Corollary 11.8, with $A = B = d_\Pi I$ gives $e_\Pi * e_\Pi = e_\Pi$. Its second case gives $e_\Pi * e_{\Pi'} = 0$ for $\Pi' \neq \Pi$. ∎

As in the Finite Groups course, the kernels e_Π are orthogonal central idempotents. Substituting $y = xg$ in the formula of the corollary gives $P_\Pi f = d_\Pi \int_K \overline{\chi_\Pi(g)}\, R_g f\, dg$. The same formula projects in any finite-dimensional unitary representation σ of K: the operator $d_\Pi \int_K \overline{\chi_\Pi(g)}\, \sigma(g)\, dg$ is the orthogonal projection onto the Π-isotypic part. For trivial Π this is the invariant-vector projector of Lecture 5.

The Π-*isotypic part* of the right regular action R is the sum of all finite-dimensional R-invariant subspaces $W \subset C(K)$ with $R|_W \cong \Pi$.

We now identify M_Π as this isotypic part. By cyclicity of the trace, $c_A(xg) = \operatorname{tr}(\Pi(g)A\Pi(x))$. Hence $R_g c_A = c_{\Pi(g)A}$. The map $A \mapsto c_A$ is injective, because $\|c_A\|^2 = d_\Pi^{-1}\|A\|_{\mathrm{HS}}^2$. Let $e_1, \dots, e_{d_\Pi}$ be an orthonormal basis of V_Π. The map $A \mapsto \Pi(g)A$ sends each column Ae_j to $\Pi(g)Ae_j$. So R on M_Π is equivalent to $\Pi^{\oplus d_\Pi}$. The jth copy of Π consists of the functions c_A with $Ae_i = 0$ for $i \neq j$.

Now let $W \subset C(K)$ be finite-dimensional and R-invariant with $R|_W \cong \Pi$. Fix $\Pi' \neq \Pi$. The map $P_{\Pi'}|_W : W \to M_{\Pi'}$ intertwines R, since $P_{\Pi'} = T_{e_{\Pi'}}$ with $e_{\Pi'}$ central. Its image is 0 or isomorphic to Π, because W is irreducible. Every irreducible subrepresentation of $M_{\Pi'} \cong \Pi'^{\oplus d_{\Pi'}}$ projects nontrivially to some summand, so Schur's lemma makes it isomorphic to Π'. Hence $P_{\Pi'}W = 0$.

So for $f \in W$ we get $S_F f = P_\Pi f$ whenever $\Pi \in F$. By the L^2 convergence statement of the Plancherel theorem, Theorem 11.4(b), $\|f - P_\Pi f\| = 0$. Both functions are continuous, so positivity gives $f = P_\Pi f \in M_\Pi$. Hence every such W lies in M_Π. Conversely, M_Π is the sum of its d_Π copies of Π. Thus M_Π is the Π-isotypic part of R, and P_Π is the orthogonal projector onto it.

Example 11.15 (Central kernels on SU(2) and the circle). Let V_m be the irreducible representation of SU(2) of dimension $m+1$, and write P_m for P_{V_m}. Let $U \in \mathrm{SU}(2)$ have eigenvalues $e^{i\theta}$ and $e^{-i\theta}$ with $0 < \theta < \pi$. Then Lecture 9 gives $\chi_m(U) = \sin((m+1)\theta)/\sin\theta$. By the isotypic projector corollary, Corollary 11.14, $P_m f = (m+1)\, f * \chi_m$.

Take the kernel $k(U) = |\operatorname{tr} U|^2 = 4\cos^2\theta$. It is central, since the trace is invariant under conjugation. Since χ_1 is real and $V_1 \otimes V_1 \cong V_0 \oplus V_2$, we get $k = \chi_0 + \chi_2$. As a check, $\sin 3\theta / \sin\theta = 4\cos^2\theta - 1$.

Characters are orthonormal, so the central kernel theorem, Theorem 11.13, gives $\kappa_0 = 1$, $\kappa_2 = \frac{1}{3}$ and $\kappa_m = 0$ otherwise. Let $f \in C(\mathrm{SU}(2))$. For $F \supseteq \{V_0, V_2\}$, the sums in that theorem equal $P_0 f + \frac{1}{3} P_2 f$, and $P_0 f$ is the constant $\int_{\mathrm{SU}(2)} f\, dU$. These sums converge to $f * k$ in L^2. Both functions are continuous, so positivity gives

$$f * |\operatorname{tr}|^2 = \int_{\mathrm{SU}(2)} f\, dU + \tfrac{1}{3} P_2 f.$$

In particular $U_{11} * k = 0$, because $U_{11} = c_{E_{11}}$ lies in M_1. At $x = I$ this reads $\int a \cdot 4(\operatorname{Re} a)^2\, dy = 0$. Left translation by $-I$ preserves dy and changes the sign of the integrand. Hence this integral vanishes, as predicted.

On $K = \mathrm{U}(1)$ every $\Pi \in \widehat{K}$ is a character χ_n given by $\chi_n(z) = z^n$ for an integer n, with $d_{\chi_n} = 1$. Here $\hat{f}(\chi_n) = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) e^{-in\theta} d\theta$ is the classical coefficient. We write $\hat{f}(n)$ for $\hat{f}(\chi_n)$ and P_n for P_{χ_n}. For $f, h \in C(\mathrm{U}(1))$, the convolution theorem, Theorem 11.7, reads $(f * h)\hat{}(n) = \hat{f}(n)\hat{h}(n)$. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), and positivity, $f * h = h * f$. The group is abelian, so every function is central, and $P_n f = f * \chi_n = \hat{f}(n) z^n$.

Self-checks

1. On $\mathrm{U}(1)$, why is the reversal in the convolution theorem, Theorem 11.7, invisible?
   Answer. Each Π is one-dimensional. The transforms are numbers, and numbers commute.
2. Let $k(U) = U_{11}$ on $\mathrm{SU}(2)$. Is T_k central?
   Answer. No. Here $k = c_{E_{11}}$ for the defining representation Π, so $\hat{k}(\Pi) = \frac{1}{2} E_{11}$ is not a scalar multiple of the identity. Hence, by the central kernel theorem, Theorem 11.13, k is not central, and so T_k is not central. To check this directly, put

   $$w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad t_\theta = \operatorname{diag}(e^{i\theta}, e^{-i\theta}).$$

   Then $k(t_\theta) = e^{i\theta}$, while $k(w t_\theta w^{-1}) = e^{-i\theta}$. These differ, for example, at $\theta = \pi/2$.
3. Compute $(\chi_\Pi * \chi_\Pi)(e)$ in two ways.
   Answer. By definition it is $\int_K \chi_\Pi(y)\chi_\Pi(y^{-1})\, dy = \int_K |\chi_\Pi|^2\, dy = 1$. By the identity $e_\Pi * e_\Pi = e_\Pi$ in the isotypic projector corollary, Corollary 11.14, $\chi_\Pi * \chi_\Pi = d_\Pi^{-1} \chi_\Pi$, whose value at e is 1.

Summary

The Fourier coefficient at Π is the operator $\hat{f}(\Pi) = \int f(g)\Pi(g^{-1})\, dg$. Peter–Weyl gives Plancherel. Hence $L^2(K)$ is the Hilbert direct sum of the spaces M_Π, and the orthogonal projection onto M_Π is $P_\Pi f = d_\Pi \operatorname{tr}(\hat{f}(\Pi)\Pi(\cdot))$. The algebraic sum of these spaces is $\mathscr{R}(K)$, which is dense in $L^2(K)$. The transform of a convolution is $(f * h)\hat{} = \hat{h}\hat{f}$. The inversion

formula holds exactly for finite coefficient sums. For general $f \in L^2(K)$, the spectral partial sums converge to f in L^2 norm. Uniform inversion needs an extra hypothesis, such as the summability condition of the uniform inversion proposition, Proposition 11.10. For compact connected K, smoothness suffices by the cited estimate in Remark 11.11. Central kernels are exactly those with scalar transforms. On $C(K)$, the isotypic projectors are $f \mapsto d_\Pi f * \chi_\Pi$.

The circle and a general compact Lie group compare as follows.

$\mathrm{U}(1)$	compact Lie group K
$\hat{f}(n) \in \mathbb{C}$	$\hat{f}(\Pi) \in \mathrm{End}(V_\Pi)$
$\|f\|^2 = \sum_n \|\hat{f}(n)\|^2$	$\|f\|^2 = \sum_\Pi d_\Pi \|\hat{f}(\Pi)\|_{\mathrm{HS}}^2$
$f = \sum_n \hat{f}(n) z^n$ (in L^2)	$f = \sum_\Pi d_\Pi \operatorname{tr}\big(\hat{f}(\Pi)\Pi(\cdot)\big)$ (in L^2)
$(f * h)\hat{}(n) = \hat{f}(n)\hat{h}(n)$	$(f * h)\hat{}(\Pi) = \hat{h}(\Pi)\hat{f}(\Pi)$

Reading

Sepanski, *Compact Lie Groups*, §3.4, for Fourier theory on compact groups. Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Chapter III, §§2–4, for representative functions and the Peter–Weyl theorem behind the Plancherel theorem, Theorem 11.4. Compare each source's transform and convolution conventions with Definitions 11.1 and 11.6 before using its formulas.